

CrackSAM 2 avec Frangi–similarité



Résultats

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SAM 1, SAM 2 et SAM 3

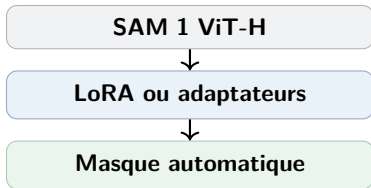


[1] Kirillov et al., ICCV 2023. **[2]** Ravi et al., 2024. **[3]** Carion et al., ICLR 2026.

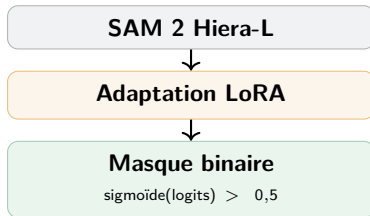


CrackSAM et baseline SAM 2

CrackSAM publié [4]



Baseline actuelle



Entraînement

Khanhha

Tests

original ; bruit 1 (assombrissement + flou 9×9) ; bruit 2 (flou 21×21 + résolution $\div 2$)

Zero-shot

Road420, Facade390, Concrete3k

CrackSAM publié

0,5780

référence externe, SAM 1

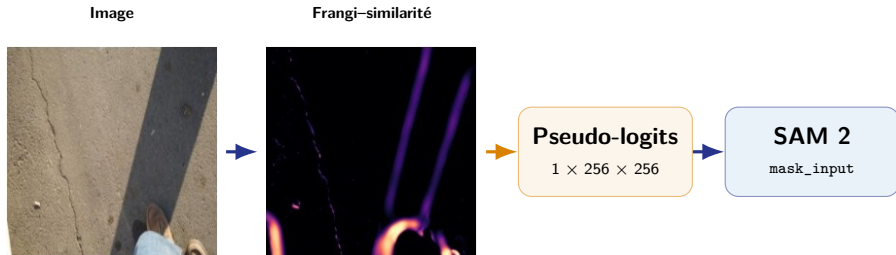
Baseline SAM 2

0,5675

IoU macro, 6 jeux



Guidage par Frangi-similarité



Baseline

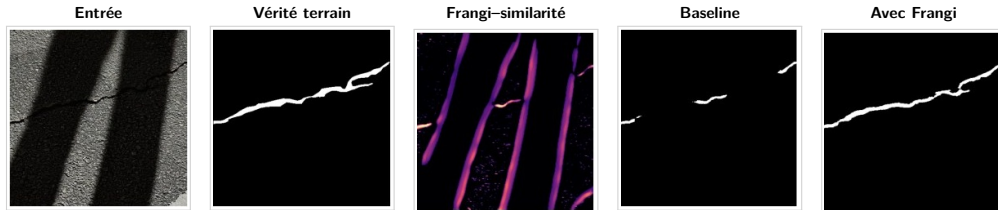
`mask_input=None`

Variante

carte Frangi sur chaque image

Même architecture et même protocole ; seul le guidage dense change.

Road420 – gain



similarité : fissure + ombres

IoU : 0,190 → 0,685

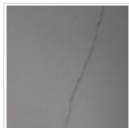
IMG_6353; $\Delta\text{IoU} = +0,4954$.



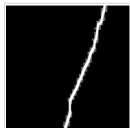
Khanhha bruité 2 et Facade390 – gains

Khanhha bruité 2

Entrée



Vérité terrain



Frangi-similarité

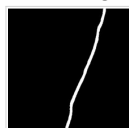


Baseline



0,000 → 0,685

Avec Frangi

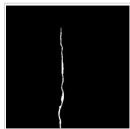


Facade390

Entrée



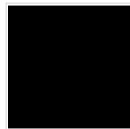
Vérité terrain



Frangi-similarité



Baseline



0,000 → 0,539

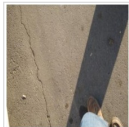
Avec Frangi



Khanhha – gains

Khanhha original

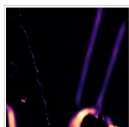
Entrée



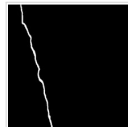
Vérité terrain



Frangi-similarité



Baseline



Avec Frangi



0,276 → 0,592

Khanhha bruité 1

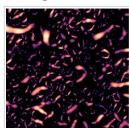
Entrée



Vérité terrain



Frangi-similarité



Baseline

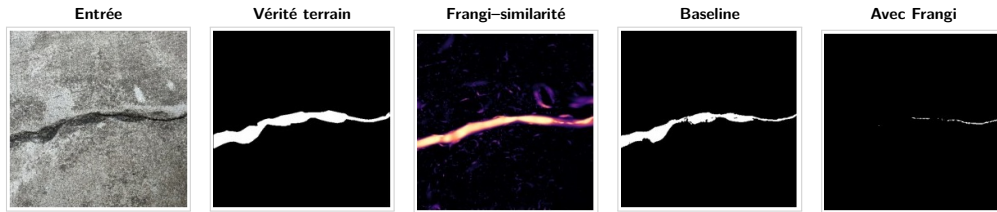


Avec Frangi



0,000 → 0,754

Road420 – perte



similarité alignée sur la fissure

IoU : 0,720 → 0,040

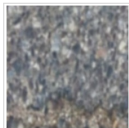
IMG_6033; $\Delta \text{IoU} = -0,6803$.



Khanhha bruité 2 et Facade390 – pertes

Khanhha bruité 2

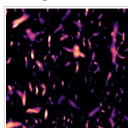
Entrée



Vérité terrain



Frangi-similarité



Baseline



0,811 → 0,000

Avec Frangi

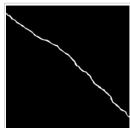


Facade390

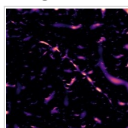
Entrée



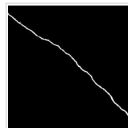
Vérité terrain



Frangi-similarité

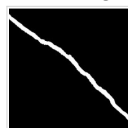


Baseline



0,750 → 0,294

Avec Frangi



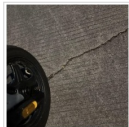


Road420 et Facade390 – autres cas

Road420 : deux bonnes prédictions

0,829 → 0,821

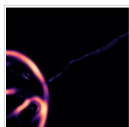
Entrée



Vérité terrain



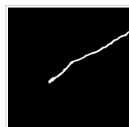
Frangi-similarité



Baseline



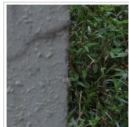
Avec Frangi



Facade390 : deux échecs

0,000 → 0,000

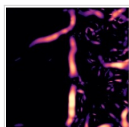
Entrée



Vérité terrain



Frangi-similarité



Baseline



Avec Frangi



Références

- [1] Kirillov et al., *Segment Anything*, ICCV 2023.
- [2] Ravi et al., *SAM 2 : Segment Anything in Images and Videos*, 2024.
- [3] Carion et al., *SAM 3 : Segment Anything with Concepts*, ICLR 2026.
- [4] Ge et al., *Fine-tuning vision foundation model for crack segmentation in civil infrastructures*, Construction and Building Materials, 2024.